

Younes Tatari

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Education

University of Utah

2023-Present

PhD in Mechanical Engineering

Thesis: Computational Modeling of Fluid-Particle Interactions: Multiphysics and Data-driven Approaches

Harbin Institute of Technology

2018-2020

MS in Energy Science and Engineering

Thesis: Numerical modeling of corona discharge using Finite Volume Method (FVM) in OpenFOAM

Skills

CFD Software OpenFOAM, SimVascular, Ansys-Fluent, Star-CCM

Fluid Physics Biotransport, Multiphase, Lagrangian particle tracking, Heat transfer, Blood flow simulation

Particle Method LIGGGHTS (CFDEM), FlowVC, DPM, MPPIC

Languages Python, Matlab, C++

Visualization ParaView, 3D Slicer

Courses FEM, Scientific Machine learning, Fluid Mechanics, Neural Networks

HPC Linux, SLURM

CAD Autodesk Inventor, SolidWorks

ML PyTorch, scikit-learn, TensorFlow, CNNs, PINNs, NumPy, Pandas, SciPy, Keras

Experience

CNH Industrial

May 2026 – Aug 2026

Soil and Crop Modeling Intern

New Holland, PA, USA

- Developed data-driven surrogate models from CFD simulations of a combine harvester cleaning system, reducing computational cost while enabling rapid performance assessment across operating conditions.

Scientific Computing and Imaging Institute at the University of Utah

Aug 2023 – present

Graduate Research Assistant

Salt Lake City, UT, USA

- Designed a novel microcatheter for targeted embolization. My design improved delivery efficiency by 10–20% compared to the standard End-Hole Catheter (SEHC) and other commercial designs. ***Provisional Patent***
- Patient Specific Modeling of Hemodynamics During Splenic Artery Embolization. 
- Implemented four-way fluid-particle coupling in successively bifurcating vessels using an Eulerian-Lagrangian CFD-DEM framework. 
- Combined Sparse Identification of Nonlinear Dynamical Systems (SINDy) with Stochastic Gradient Descent to overcome SINDy's limitation regarding prior knowledge of candidate terms in the library.
- Developed data-driven and ML to generate interpretable predictive models for fracture sealing based on CFD-DEM simulation data in geothermal systems. 

Harbin Institute of Technology

Aug 2018 – Jun 2020

Graduate Research Assistant

Harbin, China

- Investigation of cavitation under the high electric pulse in compressible fluid and electric force calculation.
- Numerical modeling of corona discharge using Finite Volume Method (FVM) in OpenFOAM.
- Facial Keypoints Detection using convolutional neural network and machine learning technics. 

Hengam company

2016–2018 & 2020 – 2022

- Designing and manufacturing solar-powered fruit dryers by leveraging CFD simulations.
- Designing and manufacturing small wind turbines to harness wind energy and generate electricity.

Publications

- A Dual-Lumen Microcatheter for Minimizing Particle Reflux During Embolization: Proof-of-concept with multiphysics simulations 2026
Computers in Biology and Medicine 
- Investigation of particle transport in geothermal systems using integrated CFD-DEM and data-driven approaches 2026
Journal of Geothermics 
- Adam-sindy: An efficient optimization framework for parameterized nonlinear dynamical system identification 2026
Journal of Physical Review Research 
- Optimizing distal and proximal splenic artery embolization with patient-specific computational fluid dynamics 2024
Journal of Biomechanics 